

2_anova_and_posthoc.R

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2026-08-08

```
# PROJECT 2: Statistical Comparison of Leaf Colours (ANOVA & Duncan Test)
# Goal: Test if plant traits significantly differ across Leaf colour groups
```

```
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.5.3
```

```
proj1 <- read_excel("C:/Users/prath/OneDrive/Desktop/proj1.xlsx",
                    col_types = c("text", "numeric", "numeric",
                                "numeric", "numeric", "text"))
```

```
d <- proj1
View(proj1)
```

```
# 1. Load Libraries
```

```
library(readxl)
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
library(agricolae)
```

```
## Warning: package 'agricolae' was built under R version 4.5.3
```

```
# 2. Fit One-Way ANOVA Model for each trait
```

```
md1 <- aov(L ~ LC, data = d)
md2 <- aov(B ~ LC, data = d)
md3 <- aov(SL ~ LC, data = d)
md4 <- aov(RL ~ LC, data = d)
summary.aov(md1)
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## LC          2   1827    913.6   107.5 <2e-16 ***
## Residuals   97    824     8.5
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary.aov(md2)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## LC              2  46.40   23.20   89.39 <2e-16 ***
## Residuals      97  25.17    0.26
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary.aov(md3)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## LC              2 1151.9   576.0   112 <2e-16 ***
## Residuals      97  498.8     5.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary.aov(md4)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## LC              2  303.5   151.74  114.3 <2e-16 ***
## Residuals      97  128.7     1.33
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 3. Post-Hoc Test: Duncan's Multiple Range Test
dr1 <- duncan.test(md1, trt="LC", group = TRUE, console = TRUE)
```

```
##
## Study: md1 ~ "LC"
##
## Duncan's new multiple range test
## for L
##
## Mean Square Error: 8.495353
##
## LC, means
##
##          L      std r      se Min Max   Q25   Q50   Q75
## Dark Green 94.2700 3.248605 40 0.4608512 88.7 99.4 91.975 93.90 97.150
## Green      88.6525 3.095322 40 0.4608512 84.7 95.7 86.350 87.95 90.650
## Light Green 82.8150 1.429124 20 0.6517420 80.6 86.2 81.775 82.80 83.725
##
## Groups according to probability of means differences and alpha level( 0.05 )
##
## Means with the same letter are not significantly different.
##
##          L groups
## Dark Green 94.2700      a
## Green      88.6525      b
## Light Green 82.8150      c
```

```
dr2 <- duncan.test(md2, trt="LC", group = TRUE, console = TRUE)
```

```
##
## Study: md2 ~ "LC"
##
## Duncan's new multiple range test
## for B
##
## Mean Square Error: 0.2595155
##
## LC, means
##
##          B      std r      se Min Max   Q25   Q50   Q75
## Dark Green 9.110 0.5865151 40 0.08054742 8.0 10.0 8.675 9.05 9.625
## Green      8.145 0.5358674 40 0.08054742 7.5  9.3 7.775 7.95 8.625
## Light Green 7.310 0.1713722 20 0.11391125 7.0  7.8 7.200 7.30 7.400
##
## Groups according to probability of means differences and alpha level( 0.05 )
##
## Means with the same letter are not significantly different.
##
##          B groups
## Dark Green 9.110      a
## Green      8.145      b
## Light Green 7.310      c
```

```
dr3 <- duncan.test(md3, trt="LC", group = TRUE, console = TRUE)
```

```
##
## Study: md3 ~ "LC"
##
## Duncan's new multiple range test
## for SL
##
## Mean Square Error: 5.142428
##
## LC, means
##
##          SL      std  r      se Min Max   Q25   Q50   Q75
## Dark Green 66.0675 2.627994 40 0.3585536 61.7 70.2 64.100 65.70 68.550
## Green      61.6475 2.308956 40 0.3585536 58.7 66.8 59.950 61.05 63.375
## Light Green 56.9600 1.064944 20 0.5070714 55.1 59.3 56.375 56.80 57.800
##
## Groups according to probability of means differences and alpha level( 0.05 )
##
## Means with the same letter are not significantly different.
##
##          SL groups
## Dark Green 66.0675      a
## Green      61.6475      b
## Light Green 56.9600      c
```

```
dr4 <- duncan.test(md4, trt="LC", group = TRUE, console = TRUE)
```

```
##
## Study: md4 ~ "LC"
##
## Duncan's new multiple range test
## for RL
##
## Mean Square Error: 1.32701
##
## LC, means
##
##           RL      std  r      se  Min  Max   Q25   Q50   Q75
## Dark Green 29.595 1.3069734 40 0.1821408 27.4 31.8 28.675 29.35 30.9
## Green      27.295 1.2112613 40 0.1821408 25.6 29.9 26.350 27.10 28.3
## Light Green 24.930 0.5068998 20 0.2575859 24.0 26.1 24.700 24.90 25.3
##
## Groups according to probability of means differences and alpha level( 0.05 )
##
## Means with the same letter are not significantly different.
##
##           RL groups
## Dark Green 29.595    a
## Green      27.295    b
## Light Green 24.930    c
```

```
# 4. Publication-Ready Boxplot
png("leaf_colour_boxplot.png", units = "in", height = 8, width = 9, res= 600)
boxplot(L ~ LC, data = d,
        main= "Boxplot of Height by Leaf colour",
        xlab = "Leaf Colour",
        ylab= "Length",
        col= "Blue", border= "Red")
dev.off()
```

```
## png
## 2
```

#5 Dark green genotypes showed the highest mean plant height.